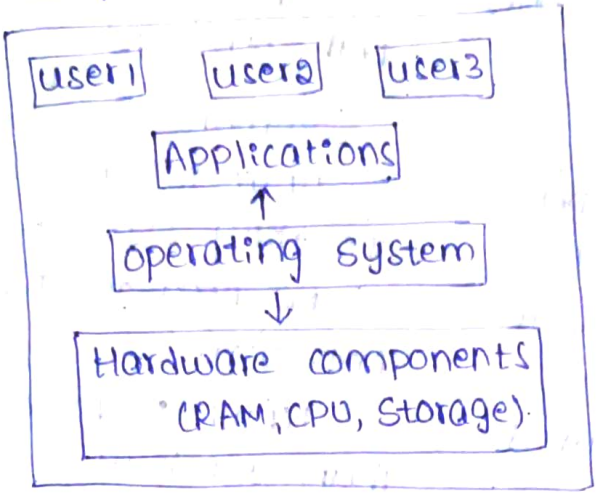
Linux

operating system: operating system is an interface b/w user and computer hardware.



Types of operating system:

They are two types of operating system

① **Single user:** only one user can login to the system at a time

Ex: windows

② **Multi user:** Multiple users can login to the system at a time

Ex: Linux

Linux: Linux is a flavour of UNIX (uniplexed information and computing system)

Ex: Red hat LINUX [RHEL], cent os, Ubuntu, SUSE Linex, Scientific Linux, Debian, Fedora, HP-UX, Solaris

→ It is an free and open-source operating system with high security. Linux is multi-user based os

Difference b/w windows and Linux :

Windows	Linux
→ windows is a single user os	→ Linux is a multi user os
→ windows is available only in GUI	→ Linux is available in both GUI and CLI (command line Interface)
→ windows is not an open source operating system	→ Linux is an open source operating system
→ Need to install Anti virus	→ No need to install any Anti virus
→ windows is less secured than LINUX	→ Linux is more secured than windows
→ windows is not a case sensitive	→ Linux is case sensitive

Minimum required to install os

RAM :: 4GB to 8GB

CPU :: 2GHz

Storage :: 25GB (Ubuntu desktop)

For practice ::

oracle virtual BOX >> Deploy Linux os

Putty

VMware work station >> Deploy Linux os

Desktop os Vs server os:

Desktop os : which we install on laptops, desktops, workstations.

Ex: windows 10, 11

ubuntu 22.4, 24

Server os : which we will install on top of servers

virtualization

Ex: windows 2022, 2025

Linux, RHEL 9.2

centos 9.0
Debian 7.0

Qs Differences b/w server os and desktop os.

Desktop os	Server os
→ Desktop os is for personal use (like browsing, gaming)	→ Server os is for managing services like websites, database, or file sharing
→ It only supports one user at a time	→ It supports multiple users or remote connections
→ It has a Graphical user interface (GUI)	→ It often runs with command line interface (CLI) only.
→ It runs apps like word, chrome, Media player	→ server os runs network services like webservers, FTP servers and databases
→ It is optimized for user experience	→ It is optimized for speed, stability and uptime
→ used by students, office workers, gamers	→ ~~comm~~ used by IT admins in data centers
Ex: windows 10/11, macos, ubuntu, desktop	Ex: windows server, ubuntu server, Red Hat, Enterprise Linux.

File operations:

To create files :: "touch"

Syntax :: touch <file_name>

Ex :: touch file1

To create multiple files ::

Ex :: touch <file> <file2> <file3> ---

List files :: "ls"

Ex :: ls <file_name>

→ To clear the screen we use the command called "clear"

Rename file :: "mv"

Ex :: mv old_file_name new_file_name

mv file1 file10

Delete file :: "rm"

→ It is to delete the files

Syntax :: rm <file1>

Ex :: rm file1

To delete multiple files ::

rm file1 file2 .... filen

To delete a file forcefully :: rm -f

Ex:: rm -f file10

Directories :

windows folders == Linux directories

To create Dir's : "mkdir"

Syntax :: mkdir <dir_name>

Ex :: mkdir dir1

Long Listing :: ls -l

Rename a directory ::

Syntax :: mv <original_dir_name> <new_dir_name>

Ex :: mv file10 file11 Ex:: mv dir1 dir10

Remove a directory ::

rm -r <dir1> # To remove directory

rm -rf <dir1> # To remove dir's forcefully

copying files :

cd :: change directory

Pwd :: present working directory

Ex :: /root/dir1

Switch directory ::

Ex :: cd dir1

cd :

cd .. :: one dir backward
cd :: Home directory of current user
cd ~ :: Home directory of current user
cd <path :: switch to the given path
cd - :: switch to previous working directory
cd / :: switch to root directory
cd ../../.. :: three steps back

copying files :: "cp" → used to copy files and directories from one place to another place
syntax :: cp source_file destination_dir
EX :: cp file1 dir4 → to copy file1 to dir4
cp file1 /root/dir3/dir4/

To copy dir's ::
syntax :: cp -r source_dir destination_dir
EX :: cp -r dir3 dir4

"ls" command:→ To list the files and directories

ls -l :: long listing
ls -S :: files with size
ls -h :: Human readable format
ls -a :: hidden files
ls -t :: Based on last modified / created (Based on timestamp)
ls -r :: To print in reverse order
ls -R :: Recursive search and list contents (print directories and subdirectories)
ls -alrt :: long listing / Reverse / Last modified / all

File Permissions:

They are three permissions in Linux

① write (w)
② Read (r)
③ Execute (x)

To check permissions
→ ls -l filename :: To get complete permission of a file

-rw-r--r-- 1 root root 0 Jul 11 9:58 file1

- rw-r--r-- → Permissions
- 1 → No.of hard links
- root → user
- root → user
- 0 → Size
- Jul 11 9:58 → timestamp
- file1 → filename

Read :: r :: 4
write :: w :: 2
Execute :: x :: 1

They are 3 types of users:
user, group, others

rw-- :: Read & write :: owner
r-- :: Read :: group
r-- :: Read :: others

→ ls -l

chmod :: change permission of a file

syntax :: chmod <permissions> filename

Ex:: chmod 222 file1 → To give write permissions

chmod 444 file1 → To give read permissions

chmod 666 file1 → To give read, write permissions

chmod 777 file1 → To give full permissions

chmod 2 file1 # chmod 002 file1

Ex:: full, read, read → # chmod 744 file1

To execute files in linux :: ./filename

chmod 777 file_name

ls -l Ex: chmod 777 ./file1

To execute this ./file_name

inode:: Inode is a data structure it contains information about files like file permissions, filename, owner, size of life

To get Inode of a file in Linux ::

ls -i <file_name> → to file inode number

ls -i file1 # 9023928

ls -i → It shows all files & directory inode numbers

ls -id ~~for~~ <dir_name> → to directory inode number

ls -id file1 # 56098909

ls -id file1 file2 # 56098909 file1 56098911 file2

Link files::

① Soft Link:: Softlink is same as windows shortcut file.

To create softlink file

Syntax:: ln -s <original_file> <softlink_file>

Ex:: ln -s file1 file2

→ Inode of original file and softlink file are different

→ If we delete original file softlink will be broken (not Accessible)

→ If we delete softlink file there will be no impact on original file

② Hard link; Hardlink is a duplicate / Backup copy for original file

To create Hardlink file

Syntax:: ln <original_file> <Hardlink_file>

Ex:: ln file1 hardlink2

→ Inode of Hardlink and original file are same

→ If we delete original file there will be no impact on hardlink file

→ If we delete Hardlink file there will be no impact on original file

→ Hardlink and original file will be in sync with other

Difference between Hard Link Vs Soft Link

Feature	Hard link	Softlink
① Points to	- file data (inode)	- file name / Path
② command to create	- ln <original_file> <Hardlink_file>	- ln -s <original_file> <softlink_file>
③ Inode number	- Same as original file	- Different inode
④ File type shown in ls -l	- Regular file	- Link (shows → pointing to file)

⑤ used for	- Duplicating file access safely	- Creating shortcuts to files or directories
⑥ can link to directories	- NO (by default)	- yes
⑦ works across file systems	- NO (same file system only)	- yes (cross file systems allowed)
⑧ Breaks if source is deleted	- NO	- yes (becomes a broken link

File operations

To add data to file / To list file contents :: cat

Syntax: cat <file_name> # To list content of a file/ To add data to a file

cat > file_name :: overwrites the data

cat >> file_name :: Append the data to existing file.

→ It creates new file / update existing file

→ we cannot edit the file

tail command :: shows the last few lines of a file (default : 10 lines)

→ To show specific number of lines

syntax :: tail -n file1 → tail -f ⇒ to get the realtime update of a file

Ex : : tail -2 file1

head command :: shows the first few lines of a file (default : 10 lines)

→ To show specific number of lines

syntax :: head -n file1

Ex :: head -2 file1

⇒ sort f1 → to sort the data in a file

It will display how many lines in that file :: wc -l file1

→ To see real time updates on file :: tail -f file1.txt used to monitor the script logs ↵

→ To see previous executed command → history (or) up & down arrows

Vi Editor (Vi → Visual Editor)

vi / vim / rvi :: used to create and edit files in Linux

→ To open file :: vi file1 [To create / update existing file]

→ "a" or "insert"

To search for keyword :: esc +:/keyword → "n" for next find

To save changes :: esc + :w

To save and exit :: esc + :wq!

To delete line :: esc + dd

To undo changes :: esc + u

To search for keyword :: esc + / keyword1

To find and replace :: esc + :%s/KW1/KW2

Switch to particular line :: esc + : line num ~~<file>~~ ~~linenum~~ Ex: esc + :4

→ esc %s/python/linux/g → All the instance
esc %s/python/linux → only at cursor position

user management commands

- To list all the users :: users
- To create new user :: useradd <username>
- To delete user :: userdel <username>
- Switch b/w users :: sudo/su <username>
- To get current user :: whoami
- To exit from current user :: exit

To change the password :: passwd <username>

To change owner of a file :: "chown"

Syntax :: chown new_ownername file_name

Ex :: chown root file1

Vi ::

End of line :: "end"

End of file :: shift + "g"

To copy paste :: select content and right click it will automatically paste

root Directory structure in Linux

/ :: Root directory [ls /]

/root :: Home directory for root user [ls /root]
↳ cd /root → ls
/home :: Home directory for other users [ls /home]
/etc :: All configuration files [ls /etc]
→ [cat /etc/passwd]
/etc/passwd :: To change/Manage password
→ [cat /etc/hostname]
/etc/hostname :: To view/change hostname
/etc/inittab [cat /etc/inittab] :: To set run level
/etc/fstab [cat /etc/fstab] :: for permanent mounting
/etc/crontab [cat /etc/crontab] :: file to schedule task
/sbin :: All dependent modules/libraries for binaries [DLL's]
/bin :: All binaries will be stored [ls /bin]
↳ cd /bin → ls
Ex :: All linux commands are binaries like ls, cp, grep

/boot :: All the boot related [os] files [ls /boot]
↳ cd /boot → ls
/dev :: Device related files [ls /dev] → cd /dev → ls
↳ cd /dev → ls
/usr :: All user related files [ls /usr]
↳ cd /usr → ls /usr/local/bin :: All local modules, binaries
/dev :: Ex :: All devices, Drives, USB, Block devices

/tmp :: All temporary files [ls /tmp]
/var :: variable files [ls /var]
/var/log [ls /var/log] → cd /var/log → ls
/var/log/boot.log [cat /var/log/boot.log] → All logs during booting process
/var/log/dmesg [cat /var/log/dmesg] → kernal logs
/var/log/messages → system event logs.

/lib :: All libraries [ls /lib]
↳ cd /lib → ls
/mnt :: All mount related files [ls /mnt]
↳ cd /mnt → ls
/proc :: All process related files [ls /proc] → cd /proc → ls
/proc/cpuinfo :: To get CPU information
/proc/meminfo :: To get memory information
[cat /proc/cpuinfo] ↓
[cat /proc/meminfo]

who -r :: To get run level

/etc/fstab
Get UUID: blkid <drive_name> or mount name
Ex: blkid /dev/sda
→ copy
vi /etc/fstab → paste :wq!

System related commands

/etc/os-release :: To get os details [cat /etc/os-release]

date :: To fetch current date

date -s "<>" :: To change date and time

man <command> :: To get manual for command
Ex :: man ls, man cp, man cat

command --help :: To get help with commands
Ex:: ls --help, etc

hostname :: To get hostname of linux machine

To change it temporarily :: hostname <new_host_name>

Ex: hostname <winteck2>

vi/etc/hostname :: To change permanently

uname :: To get system info (uname → unix name)

uname -a :: complete information
uname -r :: Kernal release
uname -m :: machine hardware name
uname -p :: processor type
uname -o :: OS
uname -v :: Kernal version
uname -s :: Kernal name
uname -n :: network node hostname
uname -i :: hardware platform

lsscsi :: To list all SCSI devices
lscpu :: To list CPU's [Site, model, Generation, core, thread]
lsmem :: To get memory information
lsblk :: To list block devices [HDD, SSD, USB, NVme, CD/DVD, Size]
RO → Rotation type
↳0 ↳1
lsusb :: To list USB (universal serial bus)
lspci :: To list all PCI Devices (peripheral component interconnect)
lshw :: To list all HW

BIOS SMBIOS: It will fetch real time hardware information from Bios and send to OS.
SMBIOS >> dmidecode
dmidecode :: system processor

dmidecode -t system (No space)
dmidecode -t memory
dmidecode -t processor
dmidecode -t baseboard
dmidecode -t bios

SMBIOS → It is read only information

↓

It is a standard that provides system hardware information like manufacturer, serial number, BIOS version, memory details

dmidecode → It will get hardware information

DMI Types: (Desktop Management Interface)

The SMBIOS specification defines the following DMI types

Type	Information
0	- BIOS [sudo dmidecode -t0]
1	- System
2	- Baseboard
3	- chassis
4	- Processor
5	- Memory controller
6	- Memory module
7	- cache
8	- Port connector
9	- System Slots
10	- on board devices
11	- OEM strings
12	- system configuration
13	- BIOS language
14	- Group Associations
15	- System Event log
16	- Physical Memory Array
17	- Memory Device
18	- 32-bit Memory Error
19	- Memory Array Mapped
20	- Memory device mapped Address
21	- Built-in pointing device
22	- Portable Battery
23	- System Reset
24	- Hardware security
25	- System power controls
26	- voltage probe
27	- cooling Device
28	- Temperature probe
29	- Electrical current Probe
30	- out-of band Remote Access
31	- Boot Integrity services
32	- System Boot
33	- 64-bit Memory Error
34	- Management Device

35 - Management Device component
36 - Management Device threshold data
37 - Memory channel
38 - IPMI Device
39 - Power supply
40 - Additional Information
41 - onboard Devices Extended Information
42 - Management Controller Host Interface

redirect data to a file ::

1. using ">>" ::

Syntax :: command >> filename
Ex :: ls >> file1 → It will transfer the data of ls related all files to file1

2. using "tee" ::

"|" :: Redirect the output of one command as a input to the next command

Ex :: ls | tee file1

grep :: used for pattern matching in Linux
to search for a keyword in a file / command output

Syntax :: grep "keyword" "file" Ex: grep "python" file1
command | grep "keyword" (or) cat file1 | grep "python"
command | grep -i "keyword"
-i :: Ignore case

Ex :: grep -i "fail" file10

dmidecode -t system | grep -i "serial number"

grep :: (Global regular expression print)

→ egrep, fgrep

egrep (extended grep) :: Ex: support expressions like |, +, ?, ().
→ dmidecode -tsystem | grep -E -i "serial | number"
→ " " " " " "s.*1"
→ "

fgrep (fixed grep) :: Ex: It looks for exact text
→ dmidecode -tsystem |grep -F -i "s.*1"
→ dmidecode -tsystem |grep -F -i "serial"

→ ls *file1 : It will come all files before file1

→ ls *fi* : It will come all files before & after fi

→ ls fi* : It will come all files after fi

→ lspci | grep -i "raid"

→ lsblk -l | grep -i "sd*"

→

System Related commands free -h :: memory information

free :: To get Total, utilized and free memory

Swap Memory :: when main memory is full system will use

df :: Disk free

du :: Disk usage

shutdown -r :: It will shutdown

Process Management

Process :: process is a instance of a application/ prgm whenever we start an application process will be started in the background

→ process will be active in background as long as the application is up and running.

PS :: To ~~list~~ get process

ps -eaf :: To list all running process

-a → all users :: Select process form all terminals/sessions

-e :: All process

-f :: To full-format listing

UID :: User ID

PID :: process ID

To search for particular process

→ ps -eaf | grep -i "bash"

→ ps -eaf | grep -i "python"

To get threads of a process :: ps -T <PID>

ps -elf

"top" :: To see live performance of linux system
uptime → To get server uptime [last rebooted]
% of CPU utilization
swap memory
RAM utilization
Running process
% Disk utilization

How to kill the process

"kill" :: To kill the process

syntax ::

kill -9 <PID> :: To kill process

pkill <Application_name>:: To kill application based on process name

Partition Management:

partition:: Dividing physical drive to multiple ~~stb~~ storage small units

500 GB → 100 GB :: partition1
100 GB :: partition2
100 GB :: partition3

To manage partitions in linux :: fdisk <Drive_name>

1 to 100GB :: /dev/sdb1

To create new partition :: fdisk <drive>

1. Select partition type

Command (m for help): n

Partition type
p primary :: we can create upto 4 partitions
e extended :: we can create more than 4

2. Select partition Number

Partition Number (2-4, default 2)

3. First sector [Default]

4. Last sector [select partition size]
   EX :: +100GB for 100GB partition.

5. Save changes :: W

To list all partitions :: fdisk -l <drive_name>

lsblk :: Look for tree structure under physical drive

To delete partitions ::

1. fdisk drive_name
2. Enter "d" to delete partition
3. select partition number
4. save changes [Enter "W"]

File system::

→ File system is a set of standards followed by OS to organize the data in a physical storage device

→ we can apply file system on top of complete physical drive or on a partition

Windows :: NTFS (New Technology file system), FAT32
Linux :: Ext3, ext4, NFS, ZFS, xfs
ESXi :: VMFS [VM file system]

Storage Array::

It has its own file system

→ Some of the storage array vendors with own file systems are: Net app
HP
DELL EMC
pure storage
DDN
Hitachi etc..

To create file system::

mkfs :: make file system

Syntax :: mkfs.ext4 <drive_name>

↓ After creating filesystem we will see in "df -h"

Ex :: mkfs.ext4 /dev/sdd

To check file system :: fsck <drive_name>

Ex:: fsck /dev/sdc

100GB :: File system : : 1 Byte :: 01010101

① Partition
② file system
③ mounting

Mounting :: Mounting storage space to the user

To mount storage : :

1. create a directory under /mnt
   Ex:: mkdir /mnt/dir1
2. Mount storage space to /mnt
   Ex:: 100 GB :: /dev/sdc1
   mount storagespace mount_dir
   mount /dev/sdc1 /mnt/dir1
3. To check mount storage space :: df -h

Mount :: 1. Local Mount
2. Remote Mount

server1 server2
100GB >>>>>> 100GB

To do remote mounting

mount -t nfs remote.server:/exported/directory /mnt/nfs-share

mount -t nfs ip_address:<storage_space> <mount dir_path>

To unmount ::

1. unmount the storage space
   umount <dir_name>
   umount /mnt/dir1
2. Delete directory
   rm -rf '/mnt/dir1

→ If we connect usb it was not showing in "this pc" so what is the alternate way to see it

(A) In disk management

Data compression / Backup:

zip :: compress and Bundle the files

Syntax :: zip <zip_file> file1 file2 file3 .... filen

Ex :: zip file1.zip file1 file2 file3

To unzip :: unzip <zip_file>

Ex:: unzip file.zip

tar :: Bundle the files

To create tar files ::

tar -cvf tar_file file1 file2 file3 --- filen

Ex:: tar -cvf file1.tar file2 file3

To extract tar file :: tar -xvf tar_file

Ex:: tar -xvf d1.tar

Log files in Linux :

/var/log :: Directory

/var/log/message1 ::

/var/log/message2 ::

/var/log/boot.log :: All logs during booting process

/var/log/crash :: crash dump / core dump

/var/log/dmesg :: a repository for device driver messages. use dmesg to see msgs in this file

System Event logs :: /var/log/messages

Kernal logs :: dmesg

Dowload a package

"wget"

wget <URL>

RPM ::

Package Management ::

→ To install / manage packages in Linux

→ packages are applications in linux

RPM :: Redhat package manager - Any Linux version
YUM :: yellowdog updater Modified - centos/RHEL
apt-get :: - ubuntu
[apt → advanced packager tool]
RPM :: offline Installation
YUM /apt-get :: online Installation
[RHEL :: Red hat enterprise Linux]

RPM::

Redhat package Manager :: offline Installation

python-3.12.rpm

Steps ::
1. Download rpm package
   RPM FIND >> website
2. Install RPM
   rpm -ivh <rpm package>
3. To list rpm packages :: rpm -qa

i → install
v → verbose (detailed info)
h → hash (#)
q → query
a → all
e → erase
u → upgrade

To uninstall rpm package
rpm -evh <RPM packagename>

To upgrade
rpm -uvh <rpm package>

apt-get :: To upgrade :: apt upgrade <package_name>

→ Mostly used in ubuntu Distribution
To install :: apt-get install <package_name>
Ex :: apt-get install fio
To remove :: apt-get remove <package_name>
To list :: apt list
apt list | grep -i "package"
apt will search for package in this repo and install package
apt repo To configure → /etc/apt/sources.list.d

YUM:: To upgrade :: yum upgrade <package_name>
To install package :: yum install <package_name>
To uninstall :: yum uninstall <package_name>
To list :: yum list
yum repo - To configure :: /etc/yum.repos.d/yum.conf

Storage Related commands:

lsblk :: To list drives
lsblk -l ::
lspci :: To list all PCI Devices
lspci | grep -i "raid"
hdparm :: To get drive details
Drive Name :: /dev/sdb
hdparm -I <Drive_name>
hdparm -I /dev/sdb :: To get detailed information about drives
S.M.A.R.T :: Self Monitoring Analysis and reporting Technology
→ To get smart data :: smartctl -a <drive_name>

Remote Management:

→ How to access System Remotely
PUTTY >> IP, username, Password
SSH :: Secure Shell >> 22
Telnet :: 23
Rconsole / Rlogin / serial console :: 513
} → Port numbers

windows to windows :: RDP >> Remote Desktop
Linux to Linux :: SSH
windows to Linux :: Putty, MPutty, Mobxtreme
→ PUTTY works based on SSH or similar protocols
Syntax :: ssh username@IP
exit :: To exit session
Inband communications :: SSh, rdp etc...
outband :: IPMI, Redfish, etc...

Network Related commands:

IP address :: To identify device in network
→ As soon as we connect ethernet cable to RJ45 connector / port
→ IP address will be assigned
→ IP address can be configured.

Types of IP::

① Static IP ::
→ we need to configure static IP
→ Static IP is fixed IP, It will not change incase of server reboot / Power cycle

② Dynamic IP ::
→ IP address will be allocated from DHCP
→ It is not a fixed one, If server reboots / Powercycle Dynamic IP will change

How to get IP address ?
1. Linux :: ifconfig
2. windows :: ipconfig

How to get MAC ?
1. Linux :: ifconfig
2. windows :: getMAC

1. IP address
2. MAC address
3. Diff b/w IP & MAC
4. what are different classes in IP
5. Static IP
6. Dynamic IP
7. what is subnet mask
8. what is default Gateway
9. what is DNS
10. what is DHCP
11. *** How Server will get IP in DHCP mode ?

12. port numbers of SSH, Telnet ?

13. IPV4 Vs IPV6

① IP address ::

→ A unique numerical address given to each device in a network.

Ex: 192.168.0.173

→ we can check ip address using command "ifconfig".

② MAC address ::

→ A hardware address permanently assigned to a network card (NIC)

→ Works like a "fingerprint" of your device

→ Ex: 00:1A:2B:3C:4D:5E

③ Diff b/w IP and MAC ::

Feature	IP address	MAC address
Type	- Logical (can change)	- physical (fixed in NIC)
Purpose	- Identifies device in network	- Identifies device uniquely at hardware level
Example	- 192.168.1.5	- 3C:52:82:4F:8A:BC

④ IP address classes (IPV4) ::

IP range is divided into classes (for different network size)

class A → 1.0.0.0 - 126.0.0.0 (Large networks)

class B → 128.0.0.0 - 191.255.0.0 (Medium networks)

class C → 192.0.0.0 - 223.255.255.0 (small networks)

class D → 224.0.0.0 - 239.255.255.255 (Multicast)

class E → 240.0.0.0 - 255.255.255.255 (Experimental)

⑦ what is subnet mask ::

→ used to divide IP address into Network part + Host part

Ex: IP : 192.168.1.10

subnet mask : 255.255.255.0

Network → 192.168.1 , Host → .10

⑧ what is default Gateway ::

→ the router's IP address in your network

→ Acts as the "door" to other network (like internet)

→ Ex: 192.168.1.1

⑨ what is DNS? (Domain Name System) ::
→ converts domain name into IP address
Ex: www.google.com → 142.250.190.78

⑩ what is DHCP? (Dynamic Host configuration protocol) ::
→ A server that automatically provides IP, subnet mask, gateway, DNS to devices
→ saves time, no need for manual configuration

⑫ port number of ssh, telnet ::
SSH :: port 22 (secure remote login)
Telnet :: port 23 (insecure remote login)

⑬ IPV4 and IPV6 ::

feature	IPV4	IPV6
Address size	- 32-bit	- 128-bit
Format	- 192.168.1.1	- 2001:0db8:85a3::8a2e:0370:7334
Header size	- Simple, smaller	- More complex, larger
Example use	- Most common today	- Future, slowly replacing IPV4

⑤ static IP ::
→ Manually assigned, doesn't change (used in servers, printers, etc)

⑥ Dynamic IP ::
→ Automatically assigned by DHCP, changes each time you connect (used by most home devices)

⑪ How server gets IP in DHCP mode ::
→ ① Device sends DHCP Discover (broadcast)
→ ② DHCP server replies with DHCP offer (available IP)
→ ③ Device sends DHCP request (accepts)
→ ④ Server confirms with DHCP ~~pack~~ ACK.
↳ This process is called DORA
(Discover → offer → Request → ACK)

Nice Index ::
Nice value :: It determines a priority of a process. It tells linux scheduler how much CPU time a process should get.
A lower nice value = higher priority
A higher nice " = lower priority

How to Assign / change IPaddress

Vi /etc/sysconfig/network-scripts/ <port_id>
Edit this file to change IP address [static IP]
After changing IP address we need to restart SSH & Network service.

netstat :: Network Statistics (network
nslookup :: It will get IPaddress from Hostname or (name server) Hostname from IP address
traceroute :: Network traces [How packets transfer from port to switch]

Ping :: To check remote IP is in network and reachable or not
Syntax :: Ping <IP address>
Ex :: Ping 192.168.0.173
→ Ping -c 5 192.168.0.173
→ How to check system is available & in the work?
(A) "Ping" → Ping IPaddress
SMART :: Smartctl -a <device_name>
hdparm -I <device_name>

To create/Manage RAID Levels

Software RAID :: "mdadm"
Hardware RAID :: "storcli", "perccli", "megasas"

To powercycle / reboot server

reboot
init 6
halt :: In between power off and on state

Sudo & su
Sudo reboot :: To execute command with root priveliges
su :: switch user
sudo ::

Io stat → Io stats

Booting process: From power on the server to till it reaches login prompt

~~① BIOS / UEFI~~

there are different phases in Booting process

① BIOS / UEFI
② MBR
③ GRUB
④ Kernal
⑤ Init
⑥ Run level programs

① BIOS ::

→ BIOS stands for Basic input output system.
→ It is a firmware it will perform post operation (Power on self test)

post :: BIOS will initialize all the hardware components in the server. If any hardware component is not coming up. It will throw the post code error

RAM failure
CPU failure
Invalid RAM configuration
cmos battery issue
os drive is not detecting

UEFI ::

→ UEFI stands for unified Extensible firmware Interface
→ UEFI is the advanced version of BIOS.
→ In the Booting process it will verify signatures of Boot loader terminal and other softwares.
→ BIOS will load MBR

② MBR :: → It contain boot loader information and partition table (details of disk partions

→ MBR stands for Master boot Record
→ BIOS will search for boot loader in MBR
→ Size of MBR is 512 bytes

446 bytes :: Primary boot loader
64 bytes :: Partition table info
2 bytes :: MBR validation check

→ MBR will be located in first sector of the portable drive
→ BIOS will be execute Boot ~~code~~ load from MBR
→ MBR will load ~~grep~~ grub

③ GRUB ::
→ GRUB stands for Grand unified bootloader
→ It is one of the default boot loader for linux distribution
→ GRUB can store multiple operating system images and during the boot process it will display screen to select one of the image and it will wait for 5 seconds for user to select, else it will boot with default image mentioned in /etc/grub.config
→ Grub will load Kernal.

④ Kernal ::
→ Kernal is a key component for any operating system
→ Kernal is responsible for loading temporary root file system and starting a process called "init".

init :: init is a parent process for all the child process to get execute. If init process is not running any other process will not start

→ Kernal acts like a mediator between os and hardware
→ Without Kernal os cannot talk to CPU, memory or any other hardware devices.
→ other routes of Kernal are :
process management
Memory management
file system management, system calls and user space management
→ Kernal also responsible for device drivers for all hardware devices
→ Kernal will load "init"

⑤ Init :: → It is the first user space process started by the Kernal & it reads configuration files (/etc/inittab, /etc/systemd/)

→ Init stands for initialization
→ without init we cannot boot into OS
→ Depend on the init level we set system will boot.

there are 6 init levels

0 → Halt (It stuck in bootloader)
1 → single user mode
2 → multi user mode (No network) (with out network, filesystem)
3 → Multi user mode with file system & root (with network)
4 → unused [OEM]
5 → Default [it contain GUI]
6 → Reboot

⑥ Run level programs :: → It define system states

→ Depend on the init level we set particular run level programs will execute. these run levels are stored inside /etc/rc0.d.
→ these run levels are required to start some of the mandatory services to work with Linux systems.

/etc/rc0.d :: Run level 0
/etc/rc1.d :: Run level 1
/etc/rc2.d :: Run level 2
/etc/rc3.d :: Run level 3
/etc/rc4.d :: Run level 4
/etc/rc5.d :: Run level 5
/etc/rc6.d :: Run level 6

Services ::

sshd
network
display

→ Services will provide system level functionality like :: network, ssh, firewall
→ Services will start from boot time without user intervention

To manage services

service <service_name> status

- service sshd status
- service sshd stop
- service sshd restart
- NOTE :: If we change IP address we need to restart SSH service

- systemctl status <service_name>
- systemctl start <service_name>
- systemctl stop <service_name>
- To list all running services

systemctl list-units --type=service --state=running

How to copy files from one machine to another machine

windows to windows ::

1. Login to remote windows machine
2. Select files [Drag & Drop], ctrl c, ctrl v

Linux to Linux ::

scp :: secure copy
cp file1 dir1
scp file1 root@IP:Target_directory
scp file1 root@192.168.0.173:/home/dir1
cp -r dir1 dir2
scp -r dir1 root@192.168.0.173:/home/dir2

windows to Linux / Linux to windows

"winscp"

works based on "SFTP" protocol

Modules ::

To get Kernal version :: uname -r
Kernel modules :: cd /lib/modules
lsmod :: To list all modules

To install modules ::

modprobe -i <module_name>
modprobe -r <module_name>

"insmod" :: insmod <module_name>

Difference b/w insmod & modprobe

modprobe :: Automatically install dependencies

insmod :: we need to manually install the dependencies

** find command ::

To search for files / file path

→ find <target_dir> -name <file_name>

find dir1 -name file1 :: search for files in dir1

find . -name file1 :: search for file1 in current dir and its sub dir

find . -type d :: search for all directories

find . -type f :: search for all the files

find . -type l :: search for all link files

find . -type f -empty :: To list all empty files

find . -size 0

find dir1 -size +100M :: To print all files > 100MB

find . -type f -empty -delete :: To find empty files and delete

find . -size +1M -delete :: To find files >1MB and delete

Job Automation :: To schedule the task

Linux :: vi /etc/crontab :: file to schedule task

→ we need to edit this file and schedule the task

```
.---------- hour (0-23)
|  .------- day of month (1-31)
|  |  .---- month (1-12) OR jan, feb, mar, apr...
|  |  |  .- day of week (0-6) (Sunday=0 or 7) OR
|  |  |  |                    sun, mon, Tue
*  *  *  *
      user-name command to be executed
```

21 10 8 * root ls -l

10 * * * root python file1.py

Directory Handling ::

→ Directory is a group of files
→ windows folder is nothing but Linux directory

rmdir → removes a directory [empty directory]
Ex :: rmdir dir2

Linux ::

→ It supports both CLI & GUI
we can't take remote session of Linux GUI
→ It is open source
→ It is case sensitive Ex :: test.py, Test.py
Both are different

→ It is more secured
→ It works based on shell (command prompt)
Ex :: Bash, Bsh, Ksh etc
shell is command line interpreter
↳ Kernal (OS & hardware)
OS (user & hardware)
→ Bash is one of the shell
→ Kernal is a key component of OS which acts like mediator blw OS and hardware

How Linux works ::

→ when we execute command on shell it will pass the command to Kernal
→ Kernal will communicate with hardware, execute command and display output in console

"cp" → copying files
Ex :: cp source_path dest_path
cp file1.txt dir2
cp file1.txt /root/dir1/dir2
cp /root/dir1/dir2/file1.txt /root/dir4

source path :: /root/dir1/dir2/test1.py
destination dir :: /root/winteck

cp /root/dir1/dir2/test1.py /root/winteck

cp -r :: copying directories
Ex:: source_dir :: /root/dir1/dir2/
target_dir :: /root/winteck

cp -r /root/dir1/dir2/ /root/winteck

Generic commands ::

clear the terminal :: "clear"
To exit current user :: "exit"
To get current date and time :: date
To change the time and date :: date -s " "
To get help of any commands :: command <--help>
ex: ls --help
To get manual of any command :: man <command>
To get server/linux machine wake up time :: "uptime
To reboot server/linux machine :: "reboot"
To get memory usage :: free
free -h
Available memory and used memory

Total memory
used memory
Available memory
swap memory

→ Zombie process ::
→ How to get priority of a process ?
→ How to increase/decrease priority of a process ?

Grep:
Ex:: cat file1 | grep "^p" → It will list all the lines starting with "p"
cat file1 | grep -i "python" → All the lines
cat file1 | grep -i "python" | wc -l
↳ No. of lines with python & PYTHON

Ex: ls | grep ".py" | wc -l
3
ls *.py | wc -l
3

Booting process::

from power on server to till it reaches login prompt:

they are 6 stages::

1. BIOS/UEFI
2. MBR/GPT
3. GRUB
4. Kernal
5. Init
6. Rean level programs

1. BIOS::

→ BIOS stands for Basic Input output system
→ BIOS is a firmware that resides inside EEPROM
→ BIOS is responsible for performing POST operation (Power on self test)
→ It will discover all the hardware components. If any hardware component is not coming up it will throw POST code error

POST code errors like:

- 0000 :: Time issue
- 0003 :: Batter low
- 000C :: RAM failed
- 000E :: Drive failed
- 8601 :: BMC not responsding
- 8102 :: USB port failed to initialize
- 00SD :: SMART error

Based on POST code we identify the issue
→ BIOS will search for Boot loader/OS in the boot device [where we install the os]

2. MBR/GPT::

→ MBR stands for Master Boot Record
→ GPT stands for GUID partition Table
→ MBR will sit inside the Boot device (in first 512 bytes of Boot device

- 446 Bytes [primary Boot loader]
- 64 Bytes [partition table info]
- 2 Bytes [MBR validation check]

③ GRUB ::

→ GRUB stands for Grand unified Bootloader
→ It is one of the Bootloader
→ It will display multiple os image. If we select image system will boot with that particular image else it will load default image from /etc/grub.config

④ Kernal ::

→ Kernal is a key component in the side os
→ It will act as a mediator b/w os and hardware
→ Kernal will load temperory root file system and it is also responsible for process management, memory management etc
→ Kernal will start a process called init with a PID of "1".
→ It will

⑤ Init ::

→ Init stands for initialization
→ Based on the init level we set system will bootup with particular run level
→ they are 7 init levels

0 → Halt
1 → Single user mode without file system
2 → multi user mode without GUI
3 → multi user mode with GUI
4 → unused
5 → Default (x11) > full multi user mode
6 → Reboot

→ we will get the init level from cat /etc/ inittab
To set init level
① init <init_level>
② Edit vi /etc/ inittab
load run level program

⑥ Run level program ::

→ Based on the init level we set particular run level prgm will be started
→ It will start all services and process required for

logging

/etc/rc.d/rc0.d
/etc/rc.d/rc1.d
/etc/rc.d/rc2.d
/etc/rc.d/rc3.d
/etc/rc.d/rc4.d
/etc/rc.d/rc5.d
/etc/rc.d/rc6.d

It will start default services required for login.

→ Diff b/w nrml user and root user

nrml user	root user
→ has limited permissions	→ has full administrative privileges
→ cannot manage other users	→ can create, delete and modify users
→ safe for daily tasks	→ high risk if commands are executed incorrectly

Creating partition in windows ::

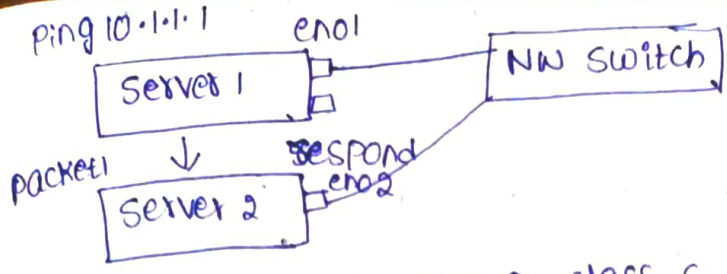
① Go to disk management
② Right click on unallocated area / diskspace
③ Next
④ select partition size
⑤ select drive letter
⑥ finish

primary drive → we will install os
secondary drive → it is for data

Packages

① yum ::

```
[base]
name = Base Repository
baseurl = http://mirror.example.com/centos/8/BaseOS/x86_64/os
enabled=1
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
```



← Ping diagram.

① what is class A, class B, class C, class D, class E

② How dynamic IP works
How DHCP works ? Assign the IPaddress

③ what is port

④ what is switch, router, hub ?

⑤ what is public & private ?
private Ex:: 10.1.1.1 , 20.1.1.1
public Ex:: 192.168.0.1, 107.111.160.12

⑥ what is VLAN ?

② DHCP (Dynamic Host configuration protocol) automatically assigns an IP address to client when it connects to the network. this avoids manual IP configuration

③ A port is a logical end point that allows multiple applications to communicate over the same IP address
Ex:: HTTP → port 80, HTTPS → port 443, SSH → port 22
FTP → port 21, DNS → port 53

④ Hub :: works at layer1 (Physical layer)
Sends data to all devices
No MAC address table

Switch :: works at layer2 (Data link layer)
uses MAC address
Sends data only to the destination device

Router :: works at layer3 (Network layer)
uses IP address
connects diff networks

⑤ public IP used for internet communication wheras private IP used within a local network
Ex: public IP :: 10.0.0.0 - 10.255.255.255
private IP :: 192.168.0.0 - 192.168.255.255
→ 49.205.15.120

⑥ VLAN is logical network that seperates devices into diff broadcast domain for better security and network management

①

class	First octet	Default subnetmask	usage
A	1-126	255.0.0.0	Large Networks
B	128-191	255.255.0.0	medium Networks
C	192-223	255.255.255.0	small Networks
D	224-239	-	multicast
E	240-255	-	experimental

① IP address
② what is class A, class B, class C, class D, class E
③ Diff b/w IPV4 Vs IPV6
④ Types of IP :: Static IP, Dynamic IP
⑤ How dynamic IP works
How DHCP works ? Assign the IPaddress
⑥ what is Subnet mask ?
⑦ what is Gateway and what is the role of default Gateway
⑧ what is DNS and what is the role of DNS in network
⑨ what is port ?
⑩ what is switch, hub, router ?
⑪ what is Private & Public IP address
⑫ what is VLAN ?

Private :: 10.0.1.1, 20.1.1.1
Public :: 192.168.0.1, 107.111.160.12

Find command : used to find the files or directories
syntax :: find <directory> -name file or dir name
<directory> :: In which directory we need to search to files or directories
"." :: search in current working directory
"/" :: search in root directory

Ex :: find/root -name file1 :: It prints all the paths of file1 in /root

find dir2 -name file2 :: It prints all the paths of file2 in dir2

find . -name file3 :: It prints all the paths of file3 in current working directory

find dir3 -type f :: It prints all the paths of a files in dir3

find / -type d :: It prints all the directories path in / directory

find /root -type l :: all the paths of a link files in /root directory

find /root -inum :: using inode number we can get path of a file

⟹ servers are pinging but ssh is failing. How to debug?

① cable connections (switch settings)
② SSH service
③ Gateway settings
   Default Gateway
④ private ip or public IP
⑤ firewall status

↳ sshd service should be up and running for ssh connections
↳ whenever we change ip address ssh service should be restarted

"*" → wildcard
zero (or) more Ex:: rm -rf *.txt, ls t*.py, cp *.py /dir1

→ metadata ?
→ device driver :: mediator ~~intermediate~~ b/w os and hardware
→ modules
   ① modprobe - Automatically install the dependencies
   ② insmod - we need to manually install the dependencies
   lsmod, /lib/modules.

Ex:- ① modprobe <module_name>
     ② insmod <module_name>

→ command to replace a string in file from command prompt

(A) sed (Stream editor)

→ can we install os without swap memory :: yes

→ Kernal modules

- KO (Kernal module)

→ command to get all empty files and delete

find . -type f -empty -delete

→ command to list all modules?

lsmod or ls /lib/modules

→ path to see system logs in linux

/var/log , /var/log/messages

/var/log/dmesg, /var/log/boot.log

→ what is swapfile and location

swapfile or swapmemory both are same - free (or) swapon

→ which memory is faster ? swap (or) main memory

main memory

→ clear installation os steps

① pendrive

② BMC

③ Pxe Boot (Network Boot)

④ Boot from SAN

① How to install os using pendrive

Step①: Download the required os ISO (ubuntu)
Step②: Create a bootable USB using tool like Rufus
Step③: Power off the server
Step④: Insert the bootable USB into servers USB port
Step⑤: Power ON the server
Step⑥: Open the Boot Menu / BIOS (Press F11/F12/Esc depending on server)
Step⑦: Select the USB device from the list of bootable devices and press Enter
Step⑧: os installer starts
Step⑨: Select installation disk. Select disk Partition or username or password
Step⑩: Click Begin Installation.
Step⑪: Wait until the installation completes
Step⑫: Remove USB drive
Step⑬: Reboot the server
Step⑭: Verify os boot successfully

② How to install os using BMC

Step①: connect your laptop to same network as the BMC
Step②: open web browser. Enter BMC / iDRAC / iLO IP Address
Ex:: https://192.168.0.xx
Step③: Login using Bmc username and password
Step④: Open virtual console / Remote console
Step⑤: Go to virtual media. Select virtual media → map CD/DVD → Browse → select os ISO → connect
Step⑥: configure one-Time Boot. Go to configuration → BIOS settings → Boot settings → one-Time Boot → Virtual CD/DVD
click Apply
Step⑦: Reboot the server
Step⑧: Server boots from mounted ISO. os installation Screen appears
Step⑨: Select installation disk. configure disk partition, hostname Root password
Step⑩: Click Begin installation and wait until it completes
Step⑪: Disconnect (unmount) the ISO from virtual media
Step⑫: Reboot the server. verify os boots successfully from installed disk

③ How to install OS using Preboot (PXE / Network Boot)

Step① : Power ON the server
Step② : Enter the Boot menu (Press F11 / F12 / ESC depend on server)
Step③ : Select Network Boot / PXE Boot / preboot Execution Environment
Step④ : Server sends a DHCP request to obtain : IP address, PXE server Address, Boot file Name
Step⑤ : DHCP server assigns an IP address and redirects the server to PXE server
Step⑥ : Server download the boot file (using TFTP) from PXE server
Step⑦ : OS installer (ubuntu) starts automatically
Step⑧ : Select Language, Timezone, Keyboard Layout
Step⑨ : Select the installation disk. Configure : hostname, password, disk partitioning
Step⑩ : Click Begin Installation. After complete reboot server
Step⑪ : Remove PXE boot option (if required) so that server boots from installed OS

④ How to install OS using Boot from SAN

Step① : Create a Boot LUN on the Hitachi Storage (EX : 100 GB)
Step② : Create a Host Group and add server's WWPN
Step③ : map the Boot LUN to Host Group
Step④ : Configure SAN zoning on the SAN switch b/w server HBA WWPN , Storage Port WWPN
Step⑤ : Verify that the server can detect the Boot LUN
Step⑥ : Open HBA Bios during server boot
Step⑦ : Configure the mapped Boot LUN as primary Boot device
Step⑧ : Mount the OS Iso using USB, Bmc Virtual media or PXE Boot
Step⑨ : Start the OS installation
Step⑩ : When prompted to select the installation disk, choose Boot LUN
Step⑪ : Complete OS installation. Reboot the server
Step⑫ : server boots using following Path.
server → HBA → SAN Switch → Hitachi Storage → Boot LUN → Operating System

Step⑬ : verify :

server boots successfully
lsblk shows the Boot LUN
multipath -ll shows active paths
OS is running from SAN LUN